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Palaeomagnetism of the Brewer Conglomerate in central Australia, and fast movement of Gondwanaland during the Late Devonian

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SUMMARY

Palaeomagnetic analysis of 11 diamond-drill cores from the Late Devonian Brewer Conglomerate in the northeastern part of the Amadeus Basin, central Australia, yields three magnetic components: a drilling-induced remanence (C1) acquired during the industrial drilling process, a mid- to Late Carboniferous syn-deformational overprint (C2) acquired during the Alice Springs Orogeny, and a primary remanence (C3). C3 passes a fold test and gives a latest Devonian pole (BC) at 47.1°S, 041.0°E with $A_{95} = 6.4°$. The revised APWP for Gondwanaland requires West Gondwanaland to drift across the South Pole during the Late Devonian with a speed of $\sim 20\,\text{cm yr}^{-1}$, while East Gondwanaland remained at low-latitude positions. It is suggested that fast plate movement occurred far more often during the Palaeozoic than has previously been thought.

Key words: Brewer Conglomerate, Devonian, drill cores, fast plate movement, Gondwanaland.

1 INTRODUCTION

Although a number of high-quality Late Devonian palaeomagnetic poles have been produced from the Lachlan Fold Belt of southeastern Australia in the last decade, the only key pole (Li & Powell 1993) for cratonic Australia is the CB pole from the Late Devonian Canning Basin reefal limestones (Hurley & Van der Voo 1987, 1990). As the position of the CB pole is over 30° away from the CV pole revealed from the Comerong Volcanics of similar age in the Tasman Fold Belt, southeastern Australia, Hurley & Van der Voo (1987) suggested that the CV pole is more likely to be erroneous because of the tectonic uncertainty of its source region. However, this view has been refuted by other workers based on either geological or palaeomagnetic arguments (e.g. Schmidt *et al.* 1986; Schmidt, Embleton & Palmer 1987; Kent & May 1987; Li, Schmidt & Embleton 1988; Li *et al.* 1990). They suggested instead that the age of the CB pole could be younger than the age of the rocks from which it was determined. This study is aimed at obtaining another key Late Devonian pole for cratonic Australia, and thereby examining the opinions above, as well as refining the palaeopositions of Gondwanaland during the Late Devonian.

Devonian rocks in the Amadeus Basin (Fig. 1a) have previously been investigated by Li *et al.* (1991a). A possible mid-Devonian primary remanence was found in only nine samples from one site in the Parke Siltstone. Another possible mid- to Late Devonian primary remanence was revealed from three out of a total of 19 sites in the Hermansburg Sandstone. Overprints related to the Alice Springs Orogeny were also found from the Late Devonian and older successions.

This study re-investigated the Brewer Conglomerate, using samples from industrial diamond drill cores from the top part of the formation—the Undandita Member, which does not crop out well at the surface.

2 GEOLOGICAL SETTING

The Brewer Conglomerate is a succession of pebble, cobble, and boulder conglomerate and subordinate interbedded sandstone and conglomeratic sandstone (Wells *et al.* 1970). Some lenses of fine-grained sandstone and black shale are present. It is the top unit of the Middle and Late Devonian Pertnjara Group (Fig. 2), and is inferred to have been deposited during an uplift associated with the early stage of the Alice Springs Orogeny (ASO) (Wells *et al.* 1970; Playford, Jones & Kemp 1976). The Undandita Member is the topmost part of the Brewer Conglomerate. It has a maximum thickness of about 1 000 m (Ferguson 1975).

The age of the Undandita Member has been determined palynologically by Playford *et al.* (1976), Balme (1988), and Evans (1988, in Jones 1991). Studies of spore assemblages indicate an age of post-early Frasnian and pre-late Famennian (Playford *et al.* 1976). Balme (1988) argued that the spore assemblages could be totally of Famennian age. More recently, a palynological study of a sample from the

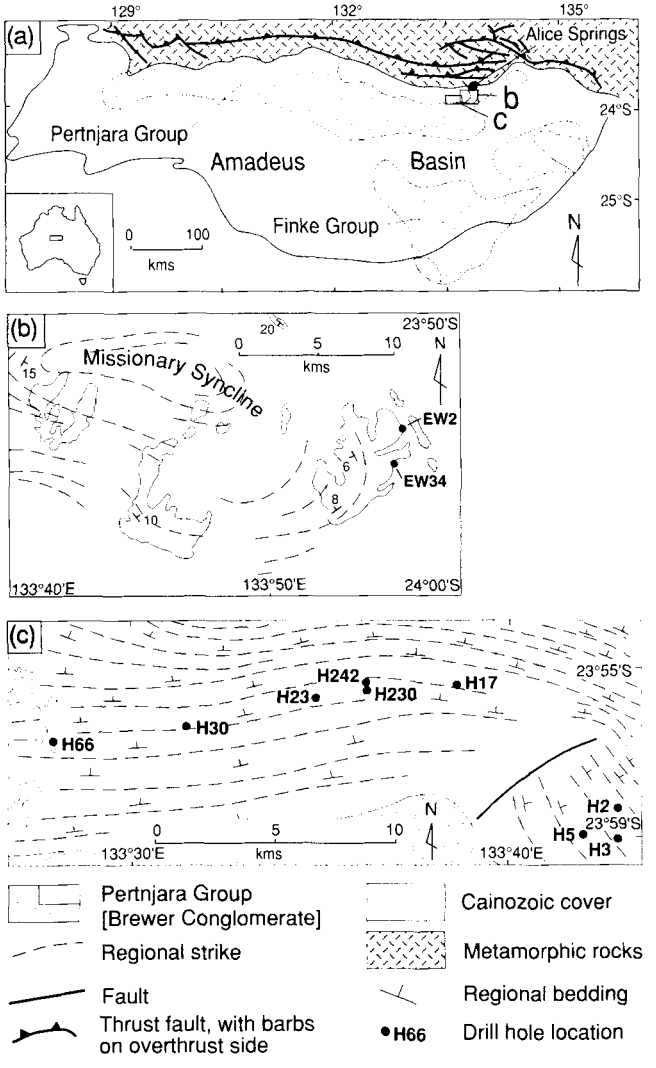


Figure 1. Regional geology and locations of the drill holes in the Amadeus Basin.

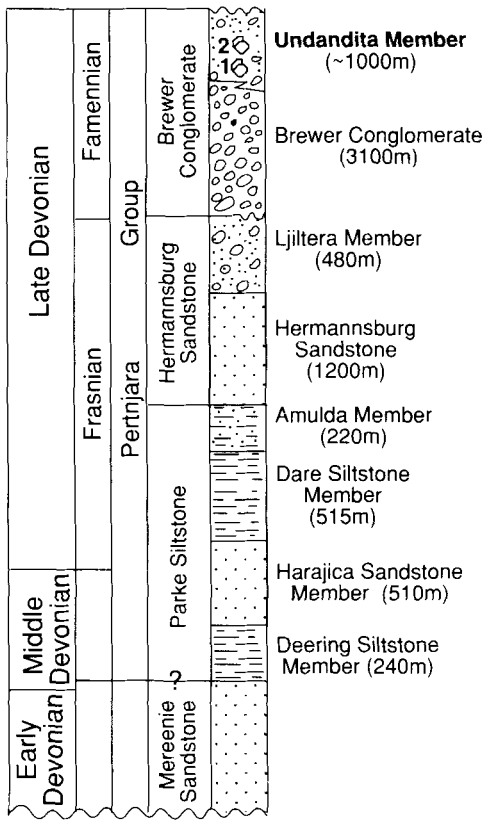


Figure 2. Stratigraphic position of the Undandita Member (modified after Fig. 2 of Playford *et al.* 1976, Jones 1991). ◇—spore assemblage, where (1) is that identified by Evans (1988; in Jones 1991), and (2) is from this study.

lower part of the Undandita Sandstone indicates a late Famennian age (Evans 1988, in Jones 1991).

In order to provide a more precise age control for the investigated materials, palynological examinations were carried out on three samples from strata encountered in drill holes (H17–62.30 m: black shale; H23–71.55 m: dark grey shale; and H66–49.60 m: grey micaceous siltstone) sunk in the middle part of the Undandita Member (Fig. 2). Well preserved and diverse plant microfossil assemblages were recovered from all three samples. They showed fairly pronounced differences in their quantitative compositions but were qualitatively similar and are of the same geological age, at least within the degree of resolution that is possible using palynology alone. They are therefore treated here for biostratigraphic purposes as a single assemblage.

The most varied assemblage was that from drill hole H23 in which over 30 distinguishable spore types were recognized. Some of these are previously undescribed forms, but others are known from the Late Devonian and Early Carboniferous of the Canning and Bonaparte Basins. They include such distinctive species as *Retusotriletes digressus* (Playford), *Knoxisporites literatus* (Waltz), *Hystricosporites porrectus* (Balme & Hassell), *Emphanisporites* sp., *Grandispora clandestina* (Playford) and *Diaphanospora* sp. cf. D. *riciniata* (Balme & Hassell).

Generally similar associations are known from the Fairfield Formation in the Canning Basin which spans the Devonian–Carboniferous boundary. This horizon is not marked by a strong palynological break but *H. porrectus*, *Emphanisporites* and *G. clandestina* were not recorded by Playford (1976) from the Tournaisian upper part of the Fairfield Formation. Their presence here therefore suggests that the samples are of uppermost Devonian age although the distinctive Famennian index species *Retispora lepidophyta* (Kedo) was not identified. This is surprising as the species is almost always present in strata of late Famennian age throughout the world and has previously been reported from a lower horizon in the Undandita Member (Playford *et al.* 1976). Its absence may perhaps be explained in terms of facies changes within the Undandita Member.

In summary, all three samples are considered to be uppermost Devonian although very close to the Carboniferous boundary. They appear to be continental as no obviously marine acritarchs were seen.

Cainozoic sediments unconformably overlie most of the Late Devonian Pertnjara Group. In the western and eastern parts of the basin, the folded Pertnjara Group outcrops

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close to the non-deformed Permian sediments (Wells *et al.* 1970). In the southeast, the Crown Point Formation, which has spore assemblages indicating Late Carboniferous to Permian age (Evans 1965, in Wells *et al.* 1970), unconformably overlies the Finke and Pertnjara Groups (Wells *et al.* 1970; Playford *et al.* 1976). These stratigraphic relationships indicate that the Late Palaeozoic deformation event in the Amadeus Basin was no later than mid-Carboniferous. This is consistent with the Rb–Sr age of 320 ± 10 Ma for the peak of the Alice Springs Orogeny (Mortimer, Cooper & James 1987; Powell & Veevers 1987).

3 SAMPLING TECHNIQUE

A total of 138 samples was collected from 11 36 mm diameter diamond-drill cores taken from the Undandita Member in the northeastern part of the Amadeus Basin by exploration companies (Table 1 and Figs. 1 & 2). Cores EW2 and EW34 were drilled near the eastern end of the Missionary Syncline (Fig. 1b), and the nine H cores were from the south limb of the Missionary Syncline (Fig. 1c). All cores were logged before sampling. Core sampling was concentrated on the reddish medium- to fine-grained sandstones and siltstones because the reddish materials were found to be more resistant to drilling-induced remanence than the greyish (reduced) rock types, and therefore more likely to retain their primary remanence (Li *et al.* 1991b).

Each sample was redrilled along the centre of the core segments to 25 mm diameter subcores in the laboratory. Two or three standard specimens of 22 mm height were then sliced from each subcore.

As all the cores were drilled vertically, orienting a sample requires only a rotation around its vertical axis until the dip-azimuth of its bedding is aligned with that of the regional bedding, which is given either by seismic data or drill-core correlation. The true dip of the bedding is the co-angle of the bedding-to-core-axis angle.

About two thirds of the specimens were processed in the palaeomagnetic laboratory of the University of Western Australia (UWA), and the rest processed in the CSIRO Rock Magnetism Laboratory at Sydney. Details of the equipment used are documented in Chen *et al.* (1993).

Table 1. Information about the studied diamond drill-cores

Core Name	Location (°N,	°E)	Bottom Depth (m)	Core Orientation	Estimated Regional Bedding (dip/dip az.)	Number of Samples
EW2	-23.90	133.92	059.51	vertical	12.0 / 250.0	07
EW34	-23.92	133.91	235.20	vertical	06.0 / 262.0	15
H2	-23.97	133.71	140.90	vertical	10.0 / 058.0	11
H3	-23.98	133.71	179.87	vertical	10.0 / 043.0	17
H5	-23.98	133.70	117.00	vertical	10.0 / 057.0	14
H17	-23.92	133.64	146.64	vertical	14.5 / 358.0	13
H23	-23.93	133.59	155.40	vertical	15.0 / 351.0	11
H30	-23.94	133.58	120.40	vertical	15.0 / 355.0	12
H66	-23.94	133.49	075.00	vertical	15.0 / 000.0	13
H230	-23.92	133.60	138.20	vertical	15.0 / 348.0	13
H242	-23.92	133.60	138.20	vertical	09.0 / 350.0	12

4 RESULTS

4.1 Anisotropy of low-field magnetic susceptibility (AMS)

Anisotropy of low-field magnetic susceptibility (AMS) of 264 specimens from 11 diamond-drill cores was measured. Specimens from cores EW2, EW34, H30 and H66 were measured on a Digico anisotropy delineator (which uses 0.7 mT, 10 Khz field) before any thermal treatment. The

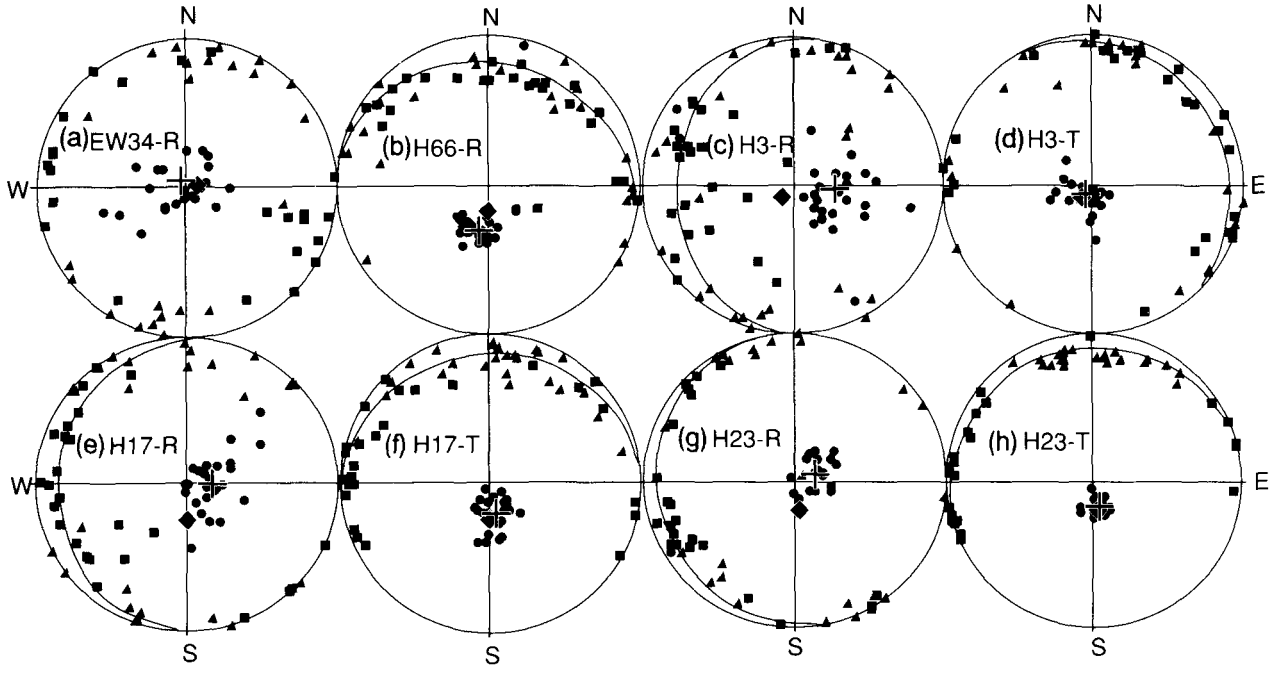


Figure 3. Representative lower hemisphere equal-area stereographic projections of the anisotropy of magnetic susceptibility (AMS) results of the drill cores of the Brewer Conglomerate. Squares represent directions of the maximum axes (K_{max}) of specimens, triangles the intermediate axes (K_{int}), dots the minimum axes (K_{min}), crosses the means of K_{min}, and diamonds the poles to the regional bedding. '$-R$' after the drill core names represents AMS data before any thermal treatment, and '$-T$' represents data after the 550 °C heating step.

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magnetic fabric is dominated by foliation, with the mean K_{min} (principal minimum axis) directions for each core close to parallel with the pole-to-bedding directions (e.g. Figs 3a and b). The bedding plane is almost flat-lying in core EW34, which makes accurate sample restoration difficult. The palaeomagnetic declinations determined from samples of core EW34 are therefore considered unreliable.

The AMS of cores H2, H3, H5, H17, H23, H230 and H242 was measured using a self-developed anisotropy system for the Bartington MS2 magnetic susceptibility meter

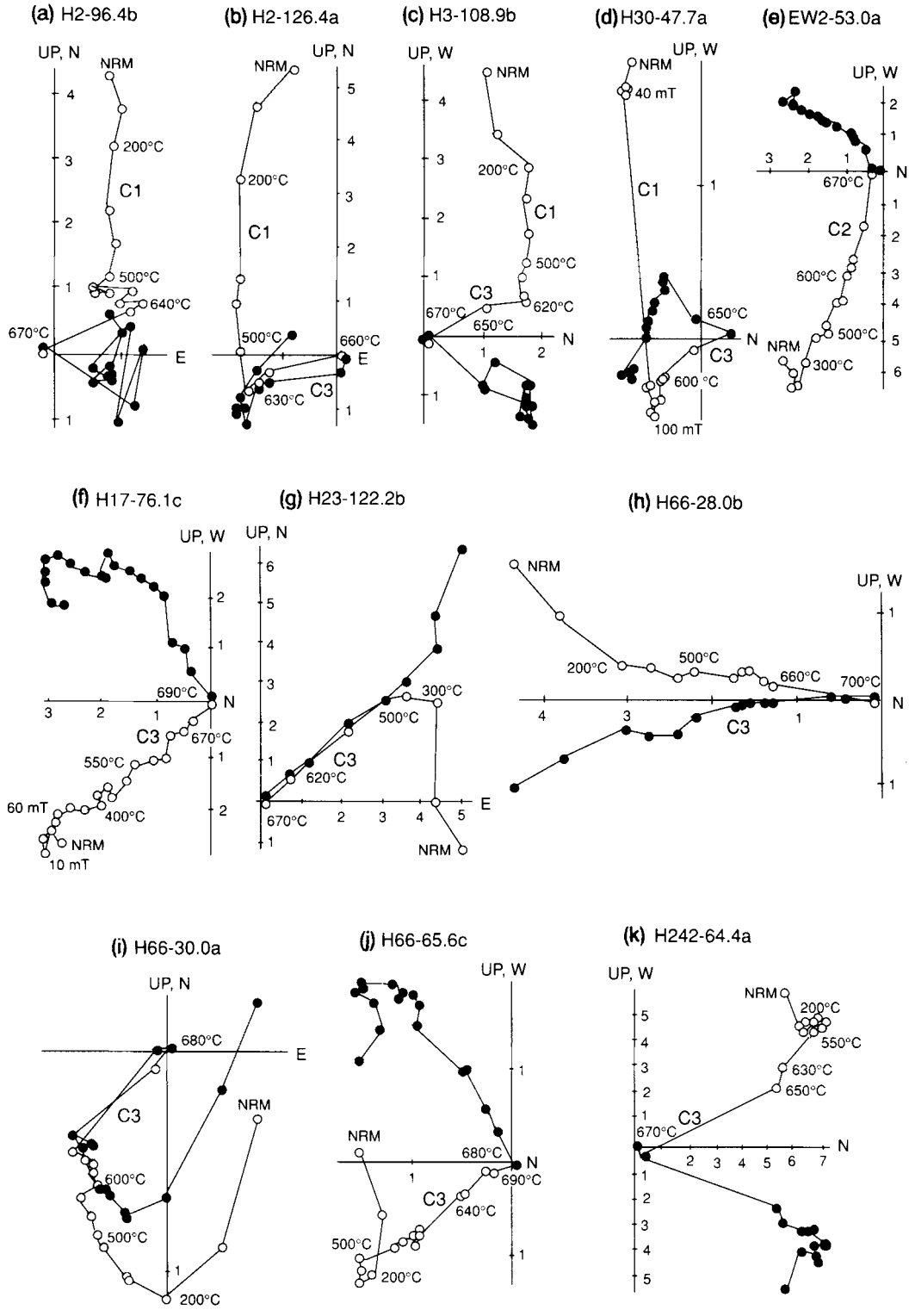


Figure 4. Representative orthogonal plots of remanent magnetization vectors. Solid (open) circles are projections on horizontal (vertical) planes of the *in situ* magnetic remanences. C1, C2 and C3 represent the identified remanent components. Scales are in $1\,mAm^{-1}$ (1×10^{-6} emu cm^{-3}).

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(which uses 0.1 mT, 0.46 or 4.6 kHz field). K_{min} directions of specimens before thermal treatment grouped reasonably well within any single core, but their mean directions were significantly different from the pole to the regional bedding (e.g. Figs 3c, e and g). However, after the 550 °C heating step, the K_{min} directions of all seven cores became better grouped, with their mean K_{min} directions falling close to the poles of the regional bedding (i.e. their magnetic foliation became close to parallel with the bedding planes) (e.g. Figs 3d, f and h). The magnetic fabric measured before thermal treatment is believed to be erronous due to the inadequate measuring positions for these low-susceptibility rocks. Details of the AMS results and their response to heating will be presented elsewhere. The well-grouped K_{min} directions indicate that the orientations for the samples, except those from core EW34, are reasonably accurate. The dominance of compactional magnetic fabric shows that the Undandita Member has not undergone any significant internal strain.

4.2 Palaeomagnetic results

NRM intensities of samples range mostly from 1.0 to 5.0 mAm^{-1} ($\times 10^{-6}$ emu cm^{-3}). All specimens were subjected to over 10 steps of thermal, or alternating field (AF) plus thermal, demagnetization. Three remanent magnetizations have been isolated from 9 out of 11 diamond-drill cores: a soft, steep-upward component (C1), an intermediate to hard component, steep-downward (C2), and a hard component, shallow-upward to the northeast, or downward to the southwest (C3). Since there is no stable component from core H230, which meets the criterion of three or more samples containing the same component with a half angle of its 95 per cent confidence cone equal to or less than 30°, this core is excluded from further analysis.

Orthogonal projections of typical demagnetization results are shown in Fig. 4. Each component is described separately below.

C1, Drilling-induced remanence (DIR)

This remanence is characterized by its steep-upward direction with low-to-medium unblocking temperature (Fig. 4a, b and c) or medium coercivity (Fig. 4d). After a principal-component analysis (Kirschvink 1980) with a linearity cut-off level of 0.98, this remanence was found in 57 per cent of the samples from 4 of the 11 diamond-drill cores (H2, H3, H5 and H30) (Table 2 and Figs 5a to d.)

The mean directions of C1 for all the four cores are closely parallel to their core axes and are significantly divergent from the present dipole field and geomagnetic field directions (Fig. 5e). They group well in geographic

Table 2. Summary of magnetic remanent directions revealed from the Brewer Conglomerate.

Core		*In Situ*		100% untilted		Common		South Pole Position			
Name	n(N)	Dec.(°)	Inc.(°)	Dec.(°)	Inc.(°)	α_{95}(°)	k	Plat. (°N)	Plong. (°E)	dp (°)	dm (°)
C1 -- drilling-induced remanence (DIR) directions											
H2	9	289.9	-85.6	253.1	-76.6	13.0	16.6				
H3	8	022.1	-87.2	230.7	-82.5	12.7	23.4				
H5	5	171.8	-82.9	210.2	-75.5	24.9	10.4				
H30	9	052.9	-76.8	119.7	-76.3	16.7	10.4				
Mean	(4)	059.3	-87.5	206.7	-82.1						
		α_{95}=9.9°	k=86.7	α_{95}=13.0°	k=51.1						
C2 -- possible mid- to Late Carboniferous syn-deformational overprint											
EW2	4	184.4	74.3	211.4	66.7	6.2	223.2				
C3 -- primary remanence											
EW2	4	224.0	41.3	227.6	30.2	08.9	107.8	-44.8	043.4	5.5	9.9
H2	5	226.2	20.3	225.1	30.3	28.5	08.2	-47.1	042.1	17.6	31.7
H3	3	034.7	-20.8	034.0	-30.7	14.9	69.1	-57.3	037.0	9.3	16.6
H17	3	227.8	22.9	233.8	31.7	27.8	20.7	-39.5	046.7	17.5	31.2
H23	3	041.5	-15.1	045.7	-24.3	28.3	20.0	-45.4	037.5	16.2	30.3
H30	5	233.0	17.6	238.1	25.0	12.1	41.1	-34.2	043.5	7.0	13.0
H66	7	220.9	18.0	225.4	28.9	23.2	07.7	-46.6	040.9	14.1	25.5
H242	3	026.5	-22.1	029.2	-29.2	20.6	36.7	-61.4	032.2	12.5	22.7
Mean											
Normal	11			039.1	-27.9	9.3	25.3				
Reverse	22			228.8	29.6	8.4	14.7				
Total	(8)	221.8	22.4	224.9	29.1			-47.1	041.0	A_{95}=6.4	
		α_{95}=7.7°	k=53.3	α_{95}=6.0°	k=86.7						

n (N) – number of samples (drill cores); dp, dm – the semiaxes of the elliptical error around a palaeopole at a probability of 95%; A_{95} (α_{95}) – half angle of the 95% confidence cone for palaeopole (mean direction); K/k – precision parameter of the Fisher distribution of palaeopoles/directions.

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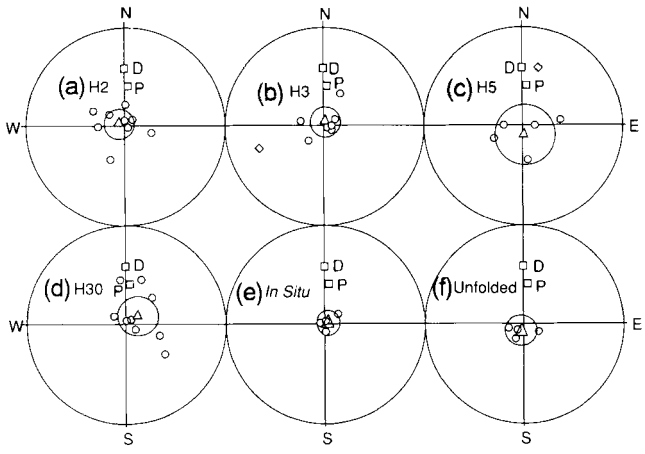


Figure 5. Upper hemisphere equal-area stereographic projections of C1 directions. (a) to (d) are *in situ* (geographic) sample-mean directions, (e) *in situ* core-mean directions and (f) core-mean directions in stratigraphic coordinates. Diamonds are the directions excluded from mean direction calculation. The means (triangles) and their α_{95} circles (small circles) are also shown for each plot. *P*—present geomagnetic field direction; *D*—present dipole field direction.

coordinates (Fig. 5e) and become slightly more dispersed in stratigraphic coordinates (Fig. 5f and Table 2).

Although core segments from core H2 were redrilled from their lower ends, and the remainder were redrilled from their upper ends during specimen preparation, the characteristics of C1 identified from core H2 are no different from those of the other three cores (Figs 4a to d, and Figs 5a to d). Furthermore, specimen sawing was done in a random manner. This means that the acquisition of C1 has no relationship with the specimen preparation in the laboratory. Therefore, this remanence is interpreted as a drilling-induced remanence (DIR) acquired during the industrial diamond drilling process, as found in samples of the Mount Eclipse Sandstone from the Ngalia Basin (Li *et al.* 1991b; Chen *et al.* 1993).

C2, possible mid- to Late Carboniferous syn-deformational overprint

This remanence can be isolated from four samples of one diamond-drill core only (EW2) (Table 2) with a linearity cut-off level of 0.98. The most important features of this component (C2) are its steep downward *in situ* direction, its medium to high unblocking temperature (300 °C to 680 °C) (Fig. 4e), and high coercivity. Samples containing this component are of reddish colour. Haematite is most likely the main magnetic carrier.

Because no fold test is available for C2, it is difficult to determine when this remanence was acquired. However, as a pre-folding remanence (see next paragraph) has been revealed from the same set of samples, C2 is unlikely to be of pre-deformational origin. On the other hand, it is unlikely to be a viscous remanence either, because its direction is different from both the present dipole field and present geomagnetic field directions (Fig. 6).

C2 has similar magnetic characteristics to a mid- to Late Carboniferous syn-deformational overprint described by Li *et al.* (1991a), and is, therefore, likely to be an overprint related to the Alice Springs Orogeny.

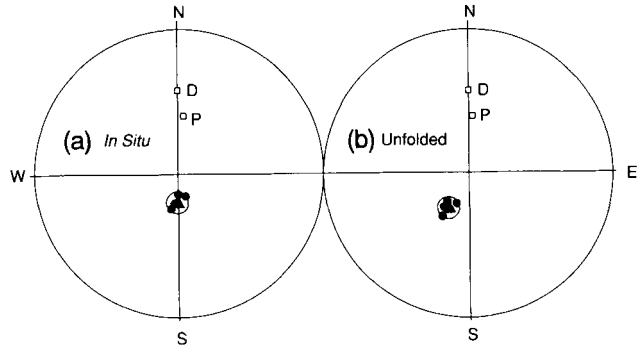


Figure 6. Lower hemisphere equal-area stereographic projections of C2 directions identified from core EW2. Symbols are the same as in Fig. 5.

C3, primary latest Devonian–earliest Carboniferous remanence

C3, which is mostly carried by haematite as suggested by its high unblocking temperature (500 °C to 690 °C) (Figs 4b, c, d, f, g, h, i, j and k) and its affinity with reddish-coloured specimens, was recognized from 8 of the 11 diamond-drill cores based on a principal-component analysis with a linearity cut-off level of 0.90. Most of the C3 vectors decay approximately towards the origin in their vector plots with their directions characterized as shallow downward to the southwest (Figs 4b, f, i and j, and Fig. 7) or upward to the northeast (Figs 4g and k, and Fig. 7) in both geographic and stratigraphic frames.

Sample-mean directions for each core are shown in Figs 7(a) to (h) and Table 2. Among the eight diamond-drill cores, data from EW2 and H66 exhibit reversed polarity (Figs 7a and g), from H23 and H242 normal polarity (Figs 7e and h), and cores H2, H3, H17 and H30 contain mixed polarities (Figs 7b, c, d and f). For cores with mixed polarities, sample-mean directions were converted to the dominant polarity before calculating the core-mean directions.

The grouping of core-mean directions of C3 was slightly improved by bedding correction (see the change in the sizes of the α_{95} circles between Figs 7i and j). The fold test of McFadden (1990) was applied to these core-mean directions (Table 3). Because the angle between the *in situ* overall mean direction (22.4°/221.8°) and the unfolded overall mean direction (29.1°/224.9°) is small (~7°), the definition-1

Table 3. Fold-test statistical parameters of the primary remanence (C3).

Remanence	N	ξ_{95}	ξ_b	ξ_a	ξ_{max}	X%	ξ_{min}	Y%
C3	8	3.298	3.631	0.787	0.804	91	0.004	096

N – number of cores; ξ_{95} – 95% critical value of test statistic ξ; $\xi_{b(a)}$ – ξ at *In Situ* (after 100% unfolding) position; ξ_{max} – ξ at maximum K position; X% – percentage of complete unfolding required for maximum K statistic; ξ_{min} – minimum ξ statistic; Y% – percentage of complete unfolding required for minimum ξ statistic.

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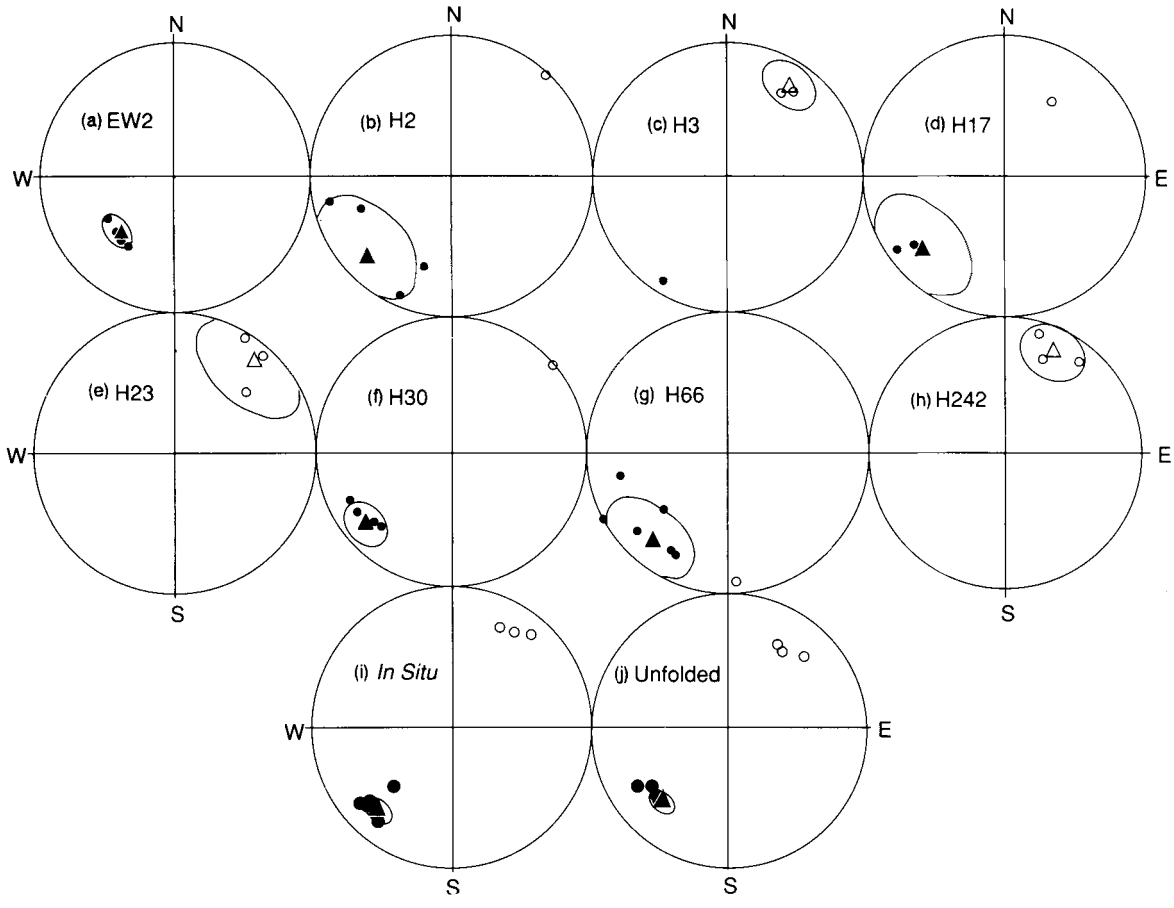


Figure 7. Equal-area stereographic projections of C3 directions. (a) to (h) are *in situ* (geographic) sample-mean directions, (i) *in situ* core-mean directions and (j) bedding-corrected (stratigraphic coordinates) core-mean directions. The means (triangles) and their α_{95} circles (small circles) are also shown. Solid (open) symbols are projections on lower (upper) hemisphere.

method of McFadden's (1990) fold test was used. The test statistic (ξ) for the *in situ* position (3.631) is higher than its 95 per cent confidence critical value (3.298), whereas for the unfolded position (0.787) it is much less than the critical value. This is a clear indication that there is correlation between the magnetic directions and the bedding orientations in the *in situ* position, but no significant correlation in the unfolded position, suggesting a pre-deformational origin for C3. The bedding-corrected sample-mean directions passed a reversal test of McFadden & McElhinny (1990) with a 'C' classification.

Figure 8 and Table 3 show that the Fisher (1953) precision parameter Kappa (κ) reaches its maximum value after 91 per cent of unfolding, whereas the fold test statistic ξ reaches its minimum value after 96 per cent of unfolding. Reasons for the different amount of unfolding required could be that: (i) most of the drilled strata have a similar strike direction and shallow dip angles (less than 15°), therefore rendering the bedding correction less definitive; (ii) the strata may have minor primary dips due to their syn-orogenic origin (Wells *et al.* 1970; Playford *et al.* 1976), and (iii) the wide variation in κ from core to core (7.7 to 107.8, see Table 2), which was not taken into account in the statistical process. Nevertheless, as the values of the half-angle of 95 per cent confidence cones (α_{95}) and κ do not change significantly during 90 per cent to 100 per cent of unfolding (Fig. 8), and the mean direction of C3 changes

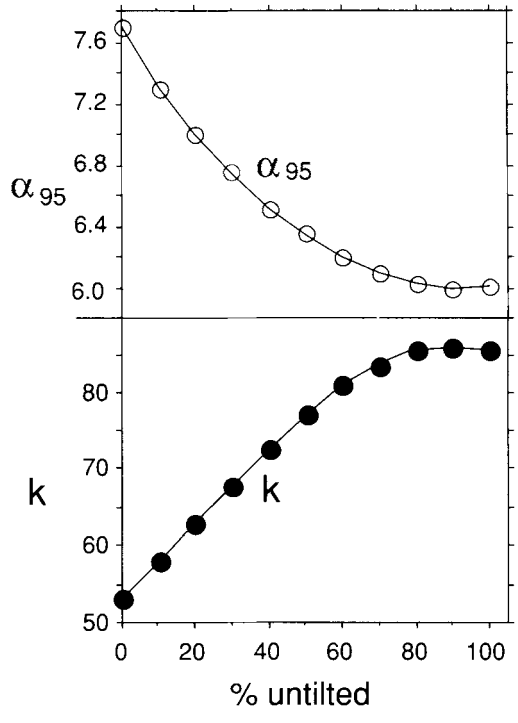


Figure 8. Changes of the Fisherian precision parameter (κ) and the half-angle of 95 per cent confidence cone (α_{95}) during stepwise unfolding of C3.

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less than 1° during the last 10 per cent of unfolding, we take the 100 per cent corrected directions as the best estimates of the directions of primary remanence. The mean of the poles calculated from core-mean directions gives a palaeo-South Pole (BC) at (47.1°S, 041.0°E) with $A_{95} = 6.4°$.

4.3 Summary of the palaeomagnetic results

A drilling-induced remanence (DIR) was revealed from segments of four diamond-drill cores. It is not as well developed as in the diamond-drill cores from the Ngalia Basin, north of the Amadeus Basin (Li *et al.* 1991b; Chen *et al.* 1993). This is probably because almost all the specimens used in this study are from reddish materials, which were reported to be more resistant to DIR (Li *et al.* 1991b). A syn-deformational overprint related to the Alice Springs Orogeny was found only in four samples from one core.

A primary magnetic remanence, which passed a positive fold test (McFadden 1990) with 95 per cent confidence, was isolated in eight drill cores. As the Undandita Sandstone, deposited during late Famennian, was deformed no later than 320 ± 10 Ma, the age of this pre-folding magnetization is confined within the latest Devonian to Early Carboniferous interval. According to Li & Powell's (1993) quality criteria, which were developed on the basis of reliability criteria of McElhinny & Embleton (1976), Briden & Duff (1981), and Van der Voo (1989, 1990), the corresponding BC pole (47.1°S, 041.0°E with $A_{95} = 6.4°$) is classified as B class. When plotted with other reliable Late Devonian–Early Carboniferous poles from Australia, it falls between the Visean MES pole (Chen *et al.* 1993) and other Late Devonian poles (Fig. 9). Its age, therefore, must be pre-Visean, i.e. late Famennian and/or Tournaisian. Our best estimate for the age of this pole is at the Devonian–Carboniferous boundary (~355 Ma, Cowie & Bassett 1989; Young & Claoué-Long 1991).

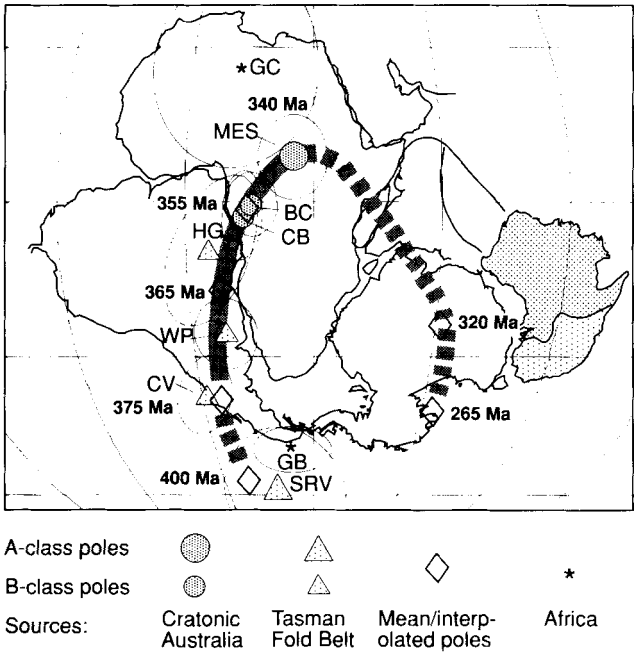


Figure 9. Revised Devonian APWP of Gondwanaland based on A- and B-class poles from Australia. The Permo-Carboniferous part of the APWP follows that of Chen *et al.* (1993). The continuous part of the path is more reliable than the dashed parts. Sources of the plotted poles are: SRV—Early Devonian Snowy River Volcanics (Schmidt *et al.* 1987); CV—early Late Devonian Comerong Volcanics (Schmidt *et al.* 1986); WP—Late Devonian Worange Point Formation (Thrupp *et al.* 1991); HG—Late Devonian Hervey Group (Li *et al.* 1988); CB—Late Devonian Canning Basin limestone (Hurley & Van der Voo 1987); BC—latest Devonian Brewer Conglomerate (this study) and MES—Early Carboniferous Mount Eclipse Sandstone (Li *et al.* 1991b; Chen *et al.* 1993); GB—originally interpreted as a late Carboniferous overprint pole from the mid-Devonian Sudan ring complexes, is reinterpreted in this study as a primary pole; GC—pole from the Sudan ring complexes, originally interpreted as primary, is reinterpreted in this study as a Devono–Carboniferous overprint. Configuration between East Antarctica, Australia and India follows Powell, Roots & Veevers (1988), and the rest follows Scotese & McKerrow (1990). Geological time scale of Cowie & Bassett (1989), with the Devonian/Carboniferous boundary at 355 Ma, has been used.

5 DISCUSSION

5.1 The Late Devonian APWP of Gondwanaland

The Devonian apparent polar wander path (APWP) for Australia is shown in Fig. 9, with circles representing poles from cratonic Australia, and triangles representing poles from the Lachlan Fold Belt, part of the Tasman Fold Belt in southeastern Australia. The Carboniferous part of the APWP follows that of Chen *et al.* (1993).

The A_{95} circle of BC pole from this study overlaps with the A_{95} circle of CB pole from the Canning Basin reefal limestones (Hurley & Van der Voo 1987) and the 95 per cent confidence oval of HG pole from the Hervey Group (Li *et al.* 1988). All these three poles are from late Frasnian–Famennian rock successions, and their positions are significantly different from the position of the MES pole of Visean age (Li *et al.* 1991b; Chen *et al.* 1993). Their ages are thus most likely the same as the rock ages. The primary origin of the CB pole is further supported by syn-depositional breccia and block tests (Chen *et al.* in preparation).

There is no geological ground for suspecting the unity of the Lachlan Fold Belt with cratonic Australia from at least the late Middle Devonian (Powell *et al.* 1990). Therefore, the difference between the CV pole (Middle/Late Devonian boundary age) and the other Late Devonian poles from Australia may have to be interpreted in term of apparent polar wandering (see discussion later). Accepting poles from the Lachlan Fold Belt as representative for Australia, we see the poles distributed linearly from the southern part of South America in the early Late Devonian, to central Africa in the Early Carboniferous (Fig. 9). The Late Devonian–Early Carboniferous segment of the Australian (and Gondwanan) APWP is thus regarded as well defined.

When the palaeomagnetic data from Australia are compared with data from other Gondwanan continents, the discrepancy with respect to the results from Africa (Bachtadse & Briden 1991) is most serious. For the purposes of the present discussion, no attempt has been made to classify the quality of Bachtadse & Briden's (1991) palaeomagnetic data.

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The trend of the Gondwanan APWP indicated from Australian data contrasts with the trend proposed by Bachtadse & Briden (1991). That is mainly because Bachtadse & Briden's (1991) 377 ± 5 Ma primary pole from the ring complexes in Sudan (pole GC) does not agree with a pole (CV) of similar age from the Comerong Volcanics in the Lachlan Fold Belt (Fig. 9; Schmidt *et al.* 1986). We believe the discrepancy might be resolved by reconsidering the chronology of the components of magnetization discussed by Bachtadse & Briden (1991). As shown in Fig. 9, the GB pole from the Sudan ring complexes, originally interpreted as a Carboniferous overprint pole, falls between the Early Devonian Snowy River Volcanics pole (SRV, Schmidt *et al.* 1987) and the early Late Devonian (~375–370 Ma) CV pole. On the other hand, the 95 per cent confidence oval of the GC pole overlaps with the A_{95} circle of the Early Carboniferous MES pole from the Mount Eclipse Sandstone (Li *et al.* 1991b; Chen *et al.* 1993). We therefore suggest that the GB pole could represent the primary pole, and hence regard the GC pole as a secondary one. We note that Bachtadse & Briden (1991) indicated that '... in the absence of decisive field tests, the timing of acquisition of the NRM is intrinsically difficult to assess'.

5.2 Rapid plate movement during the Late Devonian

Although fast plate movement, similar to rates for the Indian plate during the Late Cretaceous (10–20 cm yr^{-1}, e.g. Sclater & Fischer 1974; Klootwijk *et al.* 1992) and for Gondwanaland during the mid-Carboniferous (~20 cm yr^{-1}, e.g. Li, Powell & Schmidt 1989; Chen *et al.* 1993), has been well documented, it has generally been regarded as exceptional. However, the palaeomagnetic data shown in Fig. 9 indicate that West Gondwanaland drifted across the South Pole during the Late Devonian at a rate of about 20 cm yr^{-1}. Note that if the Harland *et al.* (1990) Time Scale had been used in this paper then the Late Devonian drift rate would have been even higher. Rapid movement for Gondwanaland is also required for the Silurian period if the 'Y' path of Morel & Irving (1978), or its modified versions (e.g. Goleby 1980; Li *et al.* 1990), are to be adopted. Therefore, rapid continental plate motion during the Palaeozoic could be more common than generally accepted.

It is unlikely that Australia drifted at the same speed because its palaeolatitude changed little during this time interval. The data imply a rapid counter-clockwise rotation of Gondwanaland around an Euler pole close to Australia

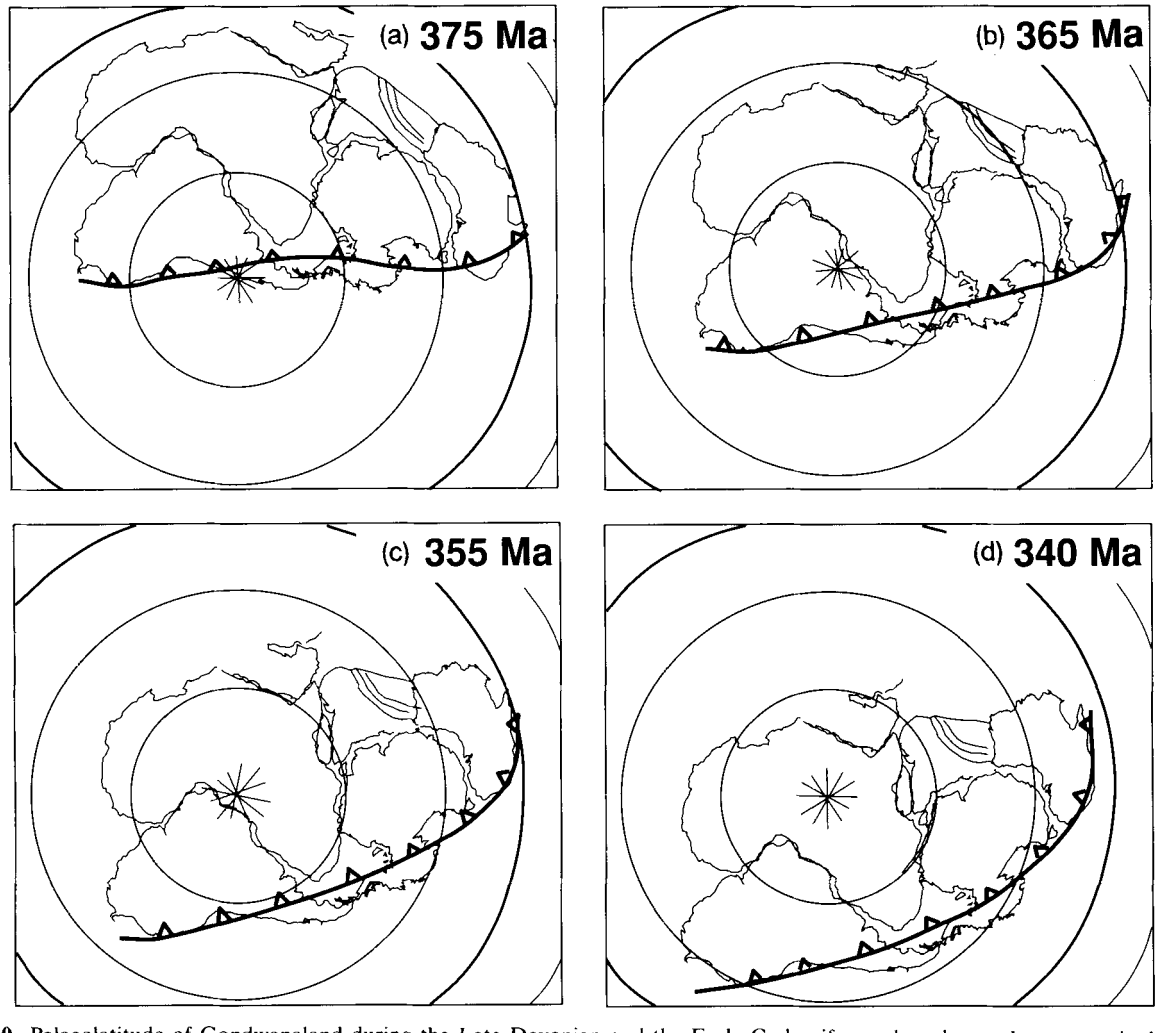


Figure 10. Palaeolatitude of Gondwanaland during the Late Devonian and the Early Carboniferous based on palaeomagnetic data shown in Fig. 9. It shows that while West Gondwanaland drifted across the South Pole, Australia remained at low-latitude positions. The trans-Gondwanan subduction zone during that time interval is marked as a saw-tooth line. Lambert equal-area projection.

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during the Late Devonian and earliest Carboniferous. Such a counter-clockwise rotation probably started as early as the Late Silurian, and lasted until earliest Carboniferous. It correlates with the dextral transtensional regime inferred to have existed in the Tasman Fold Belt of eastern Australia from mid-Silurian to mid-Devonian, followed by subduction in the Late Devonian and Early Carboniferous (Fig. 10; Powell 1984; Li *et al.* 1990). The mid-Carboniferous deformational events which marked the end of the Uluru tectonic regime (Veevers & Powell 1990) commenced as early as 340 Ma in the Lachlan Fold Belt (Powell 1984) and culminated in several tens of kilometres north–south crustal shortening in central Australia around 320 Ma (Shaw, Etheridge & Lambeck 1991). It is interesting to note that the major change in regime occurs at the same time as the sharp Early Carboniferous bend in the Gondwanan APWP.

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